

Supplementary Files

Integrative Gene Expression Analysis Using Artificial Intelligence and Network Biology Reveals Molecular Links Between Stress and Depression

Supplementary Tables:

Table S1: Complete DEG List

ID_REF	Gene Symbol	P-value	Adjusted P-value (FDR)
7896740	OR4F17 / OR4F5 / OR4F4	0.8990459	1
7896742	LOC100134822 / SEPT14 / LINC00266-1 / PCMTD2	0.881293	1
7896744	OR4F29 / OR4F21 / OR4F16 / OR4F3	0.76161	1
7896754	LOC100287934 / LOC100287497	0.6145633	1
7896756	FAM87B / FAM87A	0.8524642	1
7896759	LINC01128	0.8744782	1
7896761	SAMD11	0.1932277	1
7896779	KLHL17	0.1607697	1
7896798	PLEKHN1	0.1480021	1
7896817	ISG15	0.0670776	1
7896822	AGRN	0.4417676	1
7896859	MIR200B	0.2558359	1
7896861	MIR200A	0.1287482	1
7896863	MIR429	0.8316862	1
7896865	TTLL10	0.8297174	1

Table S2: Fisher Score ranked gene list

Feature	Fisher Score	Normalized Fisher Score
ISG15	0.03633392927721155	0.024624876499408897
MMP23A///MMP23B	0.055834713380222105	0.14003858912785747
CALML6	0.07224061812794531	0.23713552461319062
PLCH2	0.10668817931138161	0.4410104511892066

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ARHGEF16	0.04398980607944589	0.06993552454845166
HES3	0.041390810593744456	0.05455359341558649
FBXO6	0.04905831078759124	0.09993303350766923
DRAXIN	0.053077526644156284	0.12372041669241451
LOC102724562///MST1L///MST1P2///MST1	0.0608600337153994	0.16978051509177353
HTR6	0.06242900733049736	0.1790663504191119
CNKSRI	0.044565066716531994	0.07334015512660694
CATSPER4	0.03717752194512222	0.02961760711143989
OPRD1	0.045948689998286024	0.08152901045585176
TINAGL1	0.03687632419616667	0.027834994152972994

Table S3: The topological properties of the top 10 significant genes in the PPI network analyzed using Cytoscape software

Genes Name	Degree	Betweenness Centrality	Closeness Centrality	Clustering coefficient	Topological coefficient
IGF1R	59	0.8747149122807021	0.5962732919254659	5.844535359438924E-4	0.023112480739599383
STAT2	21	0.3343201754385965	0.38247011952191234	0.0	0.11904761904761904
PML	13	0.20741228070175446	0.3595505617977528	0.0	0.11538461538461539
HGF	9	0.1412280701754386	0.39834024896265563	0.027777777777777776	0.11487758945386065
HDAC1	2	0.023662280701754378	0.2981366459627329	0.0	0.5161290322580645
JAK2	2	0.08594298245614039	0.4403669724770642	0.0	0.5131578947368421
JAK1	2	0.08594298245614039	0.4403669724770642	0.0	0.5131578947368421
MDM2	2	0.16129385964912282	0.41739130434782606	0.0	0.5
ERBB2	2	0.0	0.3870967741935484	1.0	0.5151515151515151
STAT3	2	0.08594298245614039	0.4403669724770642	0.0	0.5131578947368421

Table S4: A comprehensive analysis was performed to assess the significance of the association between P-values and frequently occurring genes in the KEGG, WikiPathways, Reactome, BioCarta, BioPlanet, and DsigBD databases

Database	Term	P-value	Adjusted P-value	Genes
GO_Biological_Process	Regulation of Apoptotic Process (GO:0042981)	0.003868054783571444	0.09982545942863763	AHI1;DKKL1;PTMA;IGF1R
	Positive Regulation of Protein Localization to Chromosome,	0.00474139160939957	0.09982545942863763	PML

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	Telomeric Region (GO:1904816)			
	Cytochrome Metabolic Process (GO:1903604)	0.0047413916 0939957	0.09982545942863 763	CYP2U1
	Regulation of Protein Localization to Chromosome, Telomeric Region (GO:1904814)	0.0047413916 0939957	0.09982545942863 763	PML
	Regulation of Translational Initiation in Response to Stress (GO:0043558)	0.0056871157 38983334	0.09982545942863 763	PML
	Notch Receptor Processing (GO:0007220)	0.0066319890 6678106	0.09982545942863 763	GALNT11
	Dendritic Spine Maintenance (GO:0097062)	0.0066319890 6678106	0.09982545942863 763	IGF1R
	Negative Regulation of Vasoconstriction (GO:0045906)	0.0066319890 6678106	0.09982545942863 763	DOCK4
	Positive Regulation of Vascular Associated Smooth Muscle Cell Migration (GO:1904754)	0.0075760122 063041925	0.09982545942863 763	DOCK4
	Protein O-linked Glycosylation via Threonine (GO:0018243)	0.0085191858 00335128	0.09982545942863 763	GALNT11
GO_Cellul ar_Compo nent	Transferase Complex, Transferring Phosphorus- Containing Groups (GO:0061695)	0.0075760122 063041925	0.11622188003447 768	IGF1R

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	Primary Lysosome (GO:0005766)	0.010402987018789436	0.11622188003447768	OLFM4
	Protein Kinase Complex (GO:1902911)	0.020703445636816073	0.11622188003447768	IGF1R
	Pigment Granule (GO:0048770)	0.02256531142960272	0.11622188003447768	SYTL2
	P Granule (GO:0043186)	0.02349498640716487	0.11622188003447768	TDRD9
	Exocytic Vesicle (GO:0070382)	0.027205316711375317	0.11622188003447768	SYTL2
	Secretory Vesicle (GO:0099503)	0.02905547000861942	0.11622188003447768	SYTL2
	Melanosome (GO:0042470)	0.02905547000861942	0.11622188003447768	SYTL2
	Nucleolus (GO:0005730)	0.03892481118124631	0.1280979236750563	DOCK4;PML;IGF1R
	Nuclear Lumen (GO:0031981)	0.04003060114845509	0.1280979236750563	DOCK4;PML;IGF1R
GO_Molecular_Function	Riboflavin Transmembrane Transporter Activity (GO:0032217)	0.00474139160939957	0.09611585348385629	SLC22A14
	Semaphorin Receptor Activity (GO:0017154)	0.009461510512395743	0.09611585348385629	PLXNA2
	Neurexin Family Protein Binding (GO:0042043)	0.009461510512395743	0.09611585348385629	SYTL2
	SUMO Binding (GO:0032183)	0.013222334171232074	0.09611585348385629	PML
	Insulin-Like Growth Factor Binding (GO:0005520)	0.014160424747060764	0.09611585348385629	IGF1R
	Receptor Antagonist Activity (GO:0048019)	0.01603407239546545	0.09611585348385629	DKKL1

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	Ribosomal Small Subunit Binding (GO:0043024)	0.01790434673262116	0.09611585348385629	EIF4B
	Insulin Receptor Binding (GO:0005158)	0.01790434673262116	0.09611585348385629	IGF1R
	Steroid Hydroxylase Activity (GO:0008395)	0.021634798076404317	0.09611585348385629	CYP2U1
	Notch Binding (GO:0005112)	0.021634798076404317	0.09611585348385629	GALNT11
KEGG	Pathways in cancer	1.60E-16	1.3446270151390334e-14	HDAC1;STAT2;HGF;ERBB2;STAT3;MDM2;JAK2;PML;IGF1R;JAK1
	PI3K-Akt signaling pathway	5.83E-09	2.4488353676512615e-07	HGF;ERBB2;MDM2;JAK2;IGF1R;JAK1
	Epstein-Barr virus infection	2.42E-08	5.468676060302919e-07	HDAC1;STAT2;STAT3;MDM2;JAK1
	Proteoglycans in cancer	2.60E-08	5.468676060302919e-07	HGF;ERBB2;STAT3;MDM2;IGF1R
	Signaling pathways regulating pluripotency of stem cells	5.09E-07	8.549528886730852e-06	STAT3;JAK2;IGF1R;JAK1
	Necroptosis	7.78E-07	8.804835910119095e-06	STAT2;STAT3;JAK2;JAK1
	JAK-STAT signaling pathway	8.39E-07	8.804835910119095e-06	STAT2;STAT3;JAK2;JAK1
	Hepatitis B	8.39E-07	8.804835910119095e-06	STAT2;STAT3;JAK2;JAK1
	Influenza A	1.07E-06	9.943156465627995e-06	STAT2;JAK2;PML;JAK1
	Transcriptional misregulation in cancer	1.65E-06	1.2881310503869374e-05	HDAC1;MDM2;PML;IGF1R
WikiPathway	Cancer Pathways WP5434	1.00E-16	1.8070017059213757e-14	HDAC1;STAT2;HGF;ERBB2;STAT3;MDM2;JAK2;PML;IGF1R;JAK1

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	EGFR Tyrosine Kinase Inhibitor Resistance WP4806	9.48E-13	8.531071986554269e-11	HGF;ERBB2;STAT3;JAK2;IGF1R;JAK1
	IL6 Signaling WP364	3.85E-09	2.310978045195319e-07	HDAC1;STAT3;JAK2;JAK1
	Thyroid Stimulating Hormone TSH Signaling WP2032	1.32E-08	5.937345808511703e-07	STAT3;JAK2;IGF1R;JAK1
	IL10 Anti Inflammatory Signaling WP4495	1.98E-08	7.111172428750067e-07	STAT2;STAT3;JAK1
	ncRNAs Involved In STAT3 Signaling In Hepatocellular Carcinoma WP4337	2.57E-08	7.701780999004364e-07	STAT3;JAK2;JAK1
	Leptin Signaling WP2034	3.01E-08	7.745744117958362e-07	ERBB2;STAT3;JAK2;JAK1
	Prolactin Signaling WP2037	3.56E-08	8.020287951329207e-07	ERBB2;STAT3;JAK2;JAK1
	Immune Response To Tuberculosis WP4197	1.38E-07	2.758287372460884e-06	STAT2;JAK2;JAK1
	Androgen Receptor Network In Prostate Cancer WP2263	1.71E-07	2.992161265242917e-06	HGF;STAT3;MDM2;JAK1
Reactome	Disease	1.85E-10	5.08810374525428e-08	HDAC1;STAT2;HGF;ERBB2;STAT3;MDM2;JAK2;PML;IGF1R;JAK1
	Interleukin-20 Family Signaling	4.68E-10	6.441611160125407e-08	STAT2;STAT3;JAK2;JAK1
	Interleukin-27 Signaling	1.48E-08	1.0187898395180938e-06	STAT3;JAK2;JAK1
	Interleukin-6 Signaling	1.48E-08	1.0187898395180938e-06	STAT3;JAK2;JAK1
	Interleukin-35 Signalling	1.98E-08	1.086429121059038e-06	STAT3;JAK2;JAK1
	Diseases of Signal Transduction by Growth Factor Receptors and Second Messengers	2.47E-08	1.133955462580749e-06	HDAC1;HGF;ERBB2;STAT3;MDM2;JAK2

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	Signaling by Receptor Tyrosine Kinases	6.91E-08	2.7129193494642533e-06	HDAC1;HGF;ERBB2;STAT3;JAK2;IGF1R
	Signal Transduction	9.67E-08	3.3251481879009234e-06	HDAC1;HGF;ERBB2;STAT3;MDM2;JAK2;PML;IGF1R;JAK1
	Interferon Signaling	1.23E-07	3.4564678398350887e-06	STAT2;STAT3;JAK2;PML;JAK1
	PIP3 Activates AKT Signaling	1.26E-07	3.4564678398350887e-06	HDAC1;HGF;ERBB2;MDM2;PML
BioCarta	Role of ERBB2 in Signal Transduction and Oncology Homo sapiens h her2Pathway	6.41E-10	3.0141611886961383e-08	ERBB2;STAT3;JAK2;JAK1
	Stat3 Signaling Pathway Homo sapiens h stat3Pathway	5.03E-09	1.1828174096855429e-07	STAT3;JAK2;JAK1
	IL 6 signaling pathway Homo sapiens h il6Pathway	2.57E-08	4.022041188368946e-07	STAT3;JAK2;JAK1
	IFN gamma signaling pathway Homo sapiens h ifngPathway	3.37E-06	3.961294147616262e-05	JAK2;JAK1
	IFN alpha signaling pathway Homo sapiens h ifnaPathway	8.08E-06	7.59967793530419e-05	STAT2;JAK1
	IL22 Soluble Receptor Signaling Pathway Homo sapiens h il22bppathway	1.23E-05	9.6704003709988e-05	STAT3;JAK1
	Sumoylation by RanBP2 Regulates Transcriptional Repression Homo sapiens h ranbp2Pathway	2.04E-05	0.00013703488060306536	HDAC1;MDM2
	EGF Signaling Pathway Homo sapiens h egfPathway	4.25E-05	0.0002499536395356733	STAT3;JAK1
	TPO Signaling Pathway Homo sapiens h TPOPathway	6.17E-05	0.00032240408906845783	STAT3;JAK2

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	PDGF Signaling Pathway Homo sapiens h pdgfPathway	7.27E-05	0.0003414965125596111	STAT3;JAK1
BioPlanet	Pathways in cancer	1.95E-13	4.606233379171654e-11	HDAC1;HGF;ERBB2;STAT3;MDM2;PML;IGF1R;JAK1
	Interleukin-27-mediated signaling events	4.68E-10	5.528073577416713e-08	STAT2;STAT3;JAK2;JAK1
	Immune system signaling by interferons, interleukins, prolactin, and growth hormones	1.43E-09	1.1248071158606613e-07	STAT2;HGF;STAT3;JAK2;PML;JAK1
	Leptin signaling pathway	1.62E-08	9.568140950896422e-07	ERBB2;STAT3;JAK2;JAK1
	Thyroid-stimulating hormone signaling pathway	2.24E-08	1.0097890643139055e-06	STAT3;JAK2;IGF1R;JAK1
	Interleukin-22 soluble receptor signaling pathway	2.57E-08	1.0097890643139055e-06	STAT3;JAK2;JAK1
	Interleukin-6 signaling pathway	3.01E-08	1.015553117687874e-06	HDAC1;STAT3;JAK2;JAK1
	Prolactin activation of MAPK signaling	3.76E-08	1.1105248038169822e-06	ERBB2;STAT3;JAK2;JAK1
	Immune system	7.93E-08	2.049574689362683e-06	STAT2;HGF;STAT3;MDM2;JAK2;PML;JAK1
	STAT3 pathway	8.68E-08	2.049574689362683e-06	STAT3;JAK2;JAK1
	curcumin CTD 00000663	9.62E-10	1.0510206322410747e-06	HDAC1;HGF;ERBB2;STAT3;MDM2;JAK2;IGF1R
DSigBD	Arsenenous acid CTD 00000922	1.13E-08	6.142539063810416e-06	HDAC1;HGF;ERBB2;STAT3;MDM2;JAK2;PML;JAK1
	Capsaicin CTD 00005570	2.67E-08	9.712799933007994e-06	ERBB2;STAT3;MDM2;IGF1R;JAK1
	Vorinostat BOSS	5.68E-08	1.0055255092706986e-05	HDAC1;ERBB2;STAT3;MDM2

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	172889-27-9 BOSS	5.96E-08	1.0055255092706986e-05	ERBB2;STAT3;MDM2;JAK2
	NVP-TAE684 CTD 00004657	6.10E-08	1.0055255092706986e-05	HGF;ERBB2;STAT3
	resveratrol CTD 00002483	6.45E-08	1.0055255092706986e-05	HDAC1;HGF;ERBB2;STAT3;MDM2;PML;IGF1R;JAK1
	Erlotinib BOSS	1.11E-07	1.5187499833673485e-05	ERBB2;STAT3;JAK2;IGF1R
	podophyllotoxin TTD 00010285	2.06E-07	2.497221576453285e-05	STAT3;MDM2;JAK2
	podophyllotoxin CTD 00006577	2.93E-07	3.159426968121176e-05	STAT3;MDM2;JAK2

Table S5: WGCNA Parameters (train data)

Parameter	Value
Power (soft threshold)	6 (or sft\$powerEstimate)
Network type	Signed
TOM type	Signed
Minimum module size	30
Reassign threshold	0
Merge cut height	0.25
Numeric labels	TRUE
PAM respects dendrogram	FALSE
save TOM	TRUE

Table S6: WGCNA Module Genes

Gene Symbol	Module Color
TDRD9	Orange
OLFM4	Orange
SYTL2	Red

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PML	Blue
BTN3A2	Orange
PLXNA2	Yellow
EIF4B	Orange
CYP2U1	Orange
IGF1R	Turquoise
GALNT11	Yellow
HGF	Green
DKKL1	Green
GFI1B	Green
PTMA	Green
AHI1	Green
SCAPER	Green
SLC22A14	Green
DOCK4	Red
STAT2	Blue

Table S7: Molecular docking analysis of drugs with target genes

Gene	Drug Compounds	Binding Energy (kcal/mol)
IGF1R	3-acetylursolic acid	-7.1
	Luteolin	-6.8
	Quercetin	-6.7
	Apigenin	-6.5
STAT2	3-acetylursolic acid	-8.6
	Luteolin	-7.7
	Quercetin	-7.7
	Apigenin	-7.5
PML	3-acetylursolic acid	-7.5
	Luteolin	-7.2
	Quercetin	-7.4
	Apigenin	-7.1
HGF	3-acetylursolic acid	-8.2
	Luteolin	-8.4
	Quercetin	-8.2
	Apigenin	-7.8
HDAC1	3-acetylursolic acid	-6.2

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	Luteolin	-6.2
	Quercetin	-6.1
	Apigenin	-5.9
JAK2	3-acetylursolic acid	-10.4
	Luteolin	-8.4
	Quercetin	-8.3
	Apigenin	-8.5
JAK1	3-acetylursolic acid	-8.6
	Luteolin	-8.2
	Quercetin	-8.2
	Apigenin	-7.9
MDM2	3-acetylursolic acid	-7.8
	Luteolin	-7.0
	Quercetin	-7.1
	Apigenin	-7.1
ERBB2	3-acetylursolic acid	-8.4
	Luteolin	-7.4
	Quercetin	-7.3
	Apigenin	-6.9
STAT3	3-acetylursolic acid	-8.6
	Luteolin	-7.8
	Quercetin	-7.6
	Apigenin	-7.5

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Supplementary Figures:

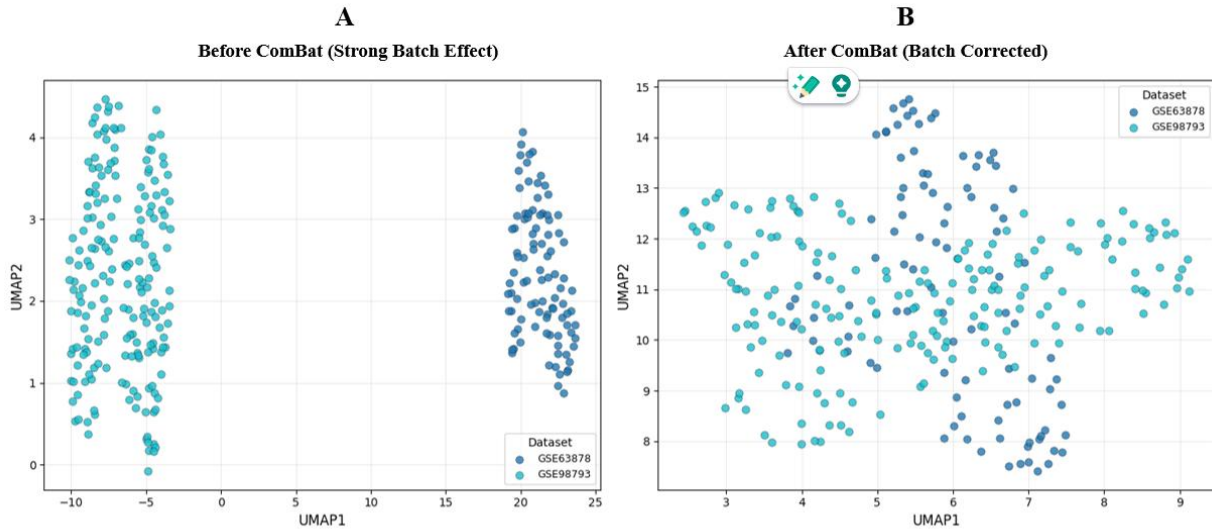


Fig. S1: Principal component analysis (UMAP) of the data before and after ComBat batch correction. A) UMAP before ComBat correction showing obvious dataset-specific clustering driven by batch effects. B) UMAP after ComBat correction illustrating significant removal of batch-associated variation while retaining biological data structure across all samples.

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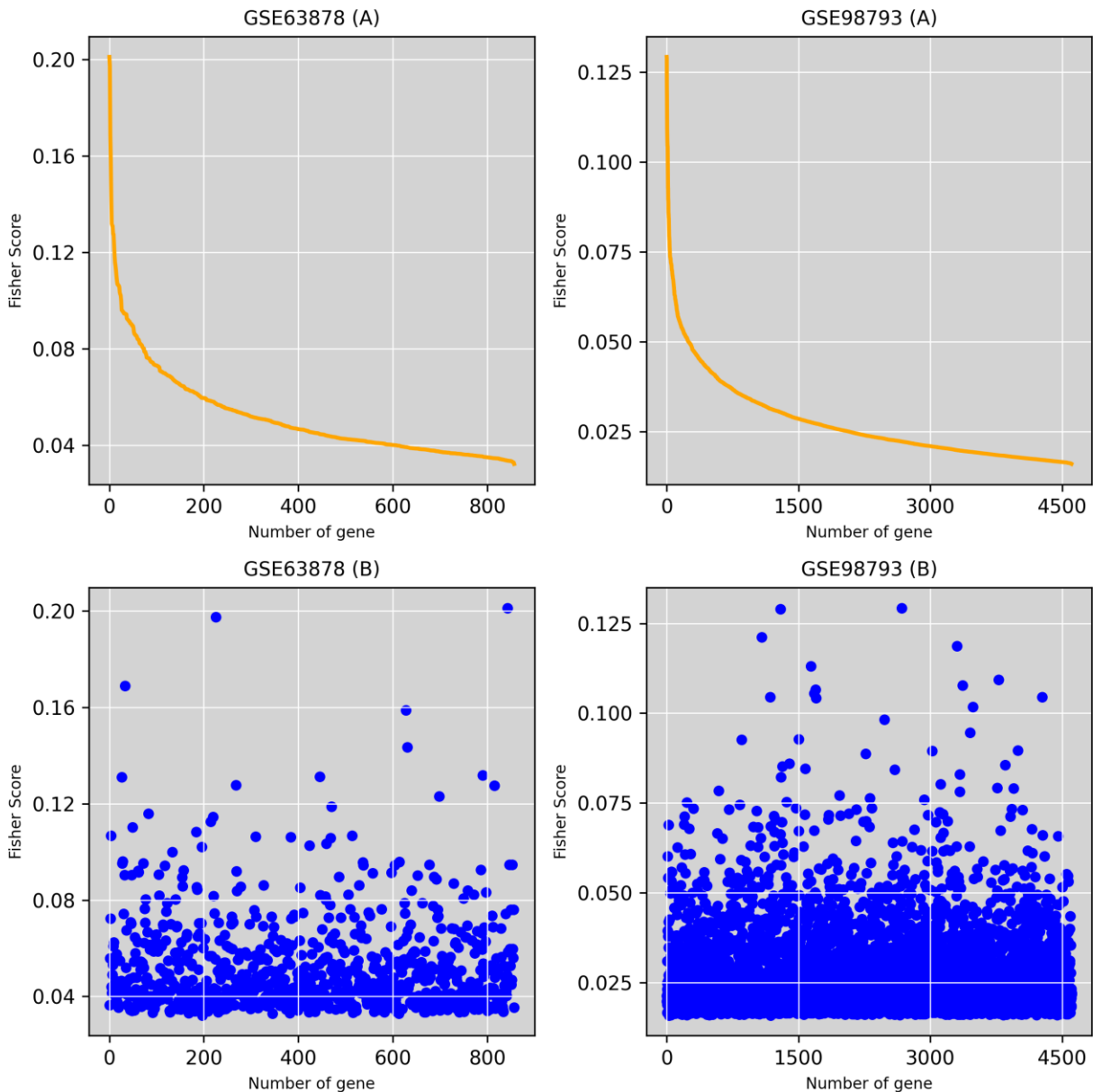


Fig. S2: The distribution of Fisher scores of genes from the depression dataset (GSE98793) and the stress dataset (GSE63878). (A) Ranked Fisher Score curves of genes from each dataset in descending order of their importance. (B) Original Fisher Score distribution curves showing the varying discriminative potential of genes from case and control samples. A plateau is evident from the ranking curve of genes, indicating that only the top 800 genes were considered.

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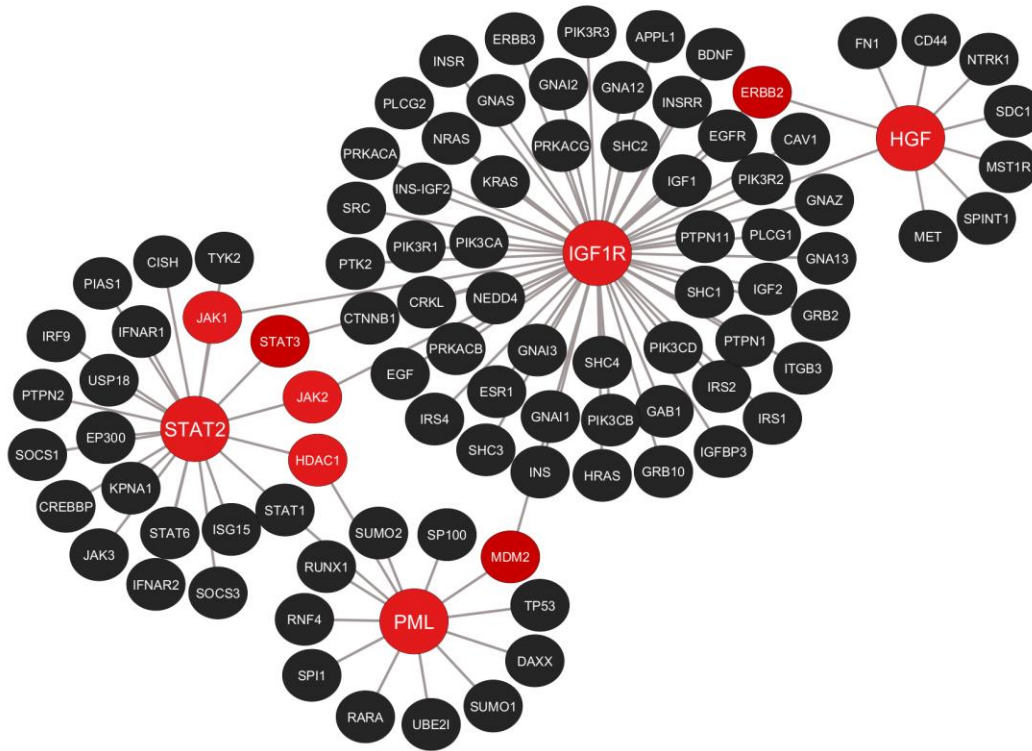


Fig. S3: Shows the PPI network generated by overlapping DEGs found in depression and stress. Red nodes signify the hub genes; black nodes indicate connections between hub genes and other connections. The PPI network had 97 nodes, 101 edges, and 4 seed genes, indicating the relationship between molecular elements and their contribution to the pathophysiology of depression and stress.

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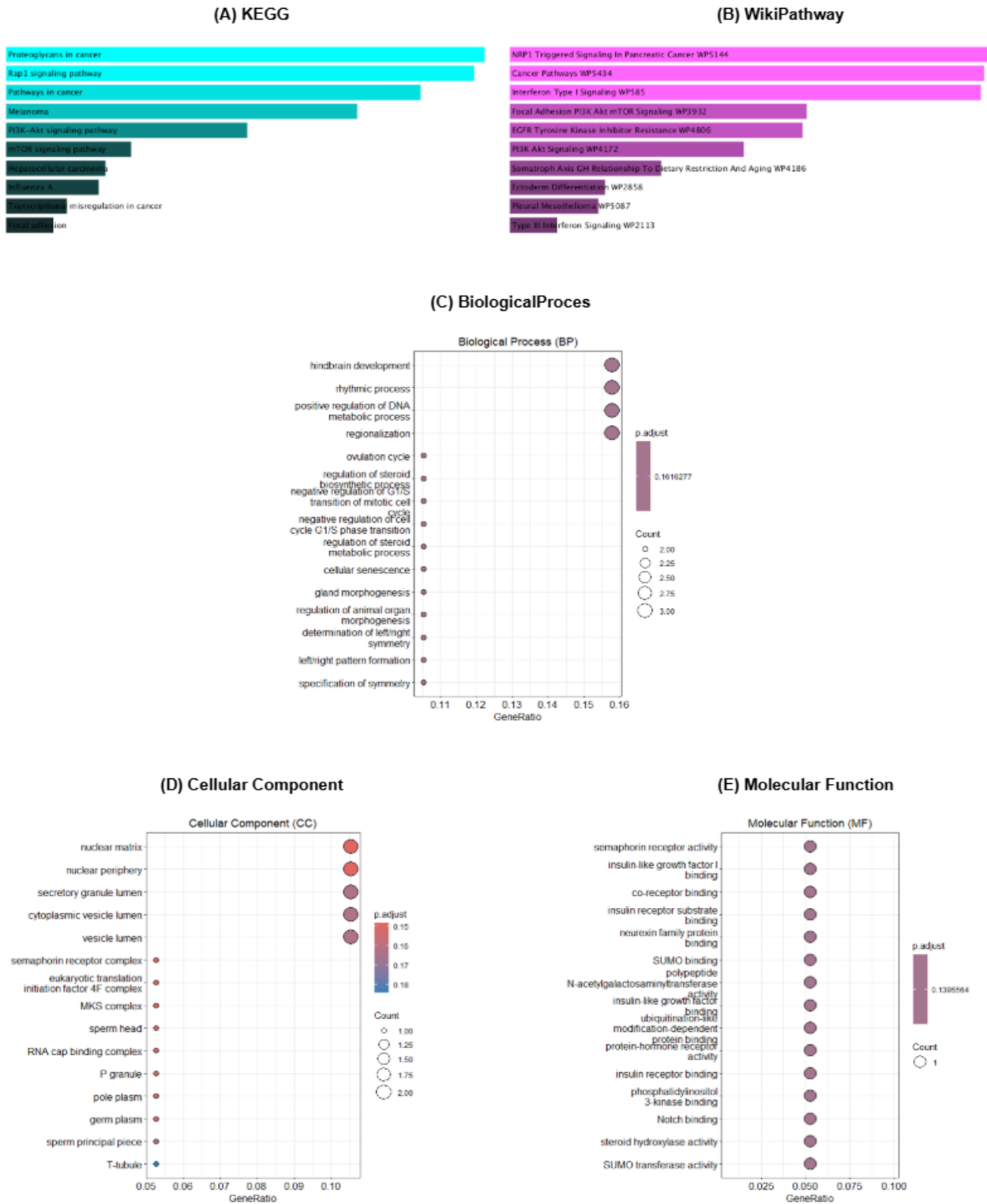


Fig. S4: Functional enrichment analysis of DEGs in depression and stress: The bar graphs from KEGG (A) and WikiPathways (B) show the main pathways both conditions share. The GO analysis - dot plots highlight

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significant biological processes (C), cellular components (D), and molecular functions (E), which provide insights into shared molecular mechanisms in depression and stress.

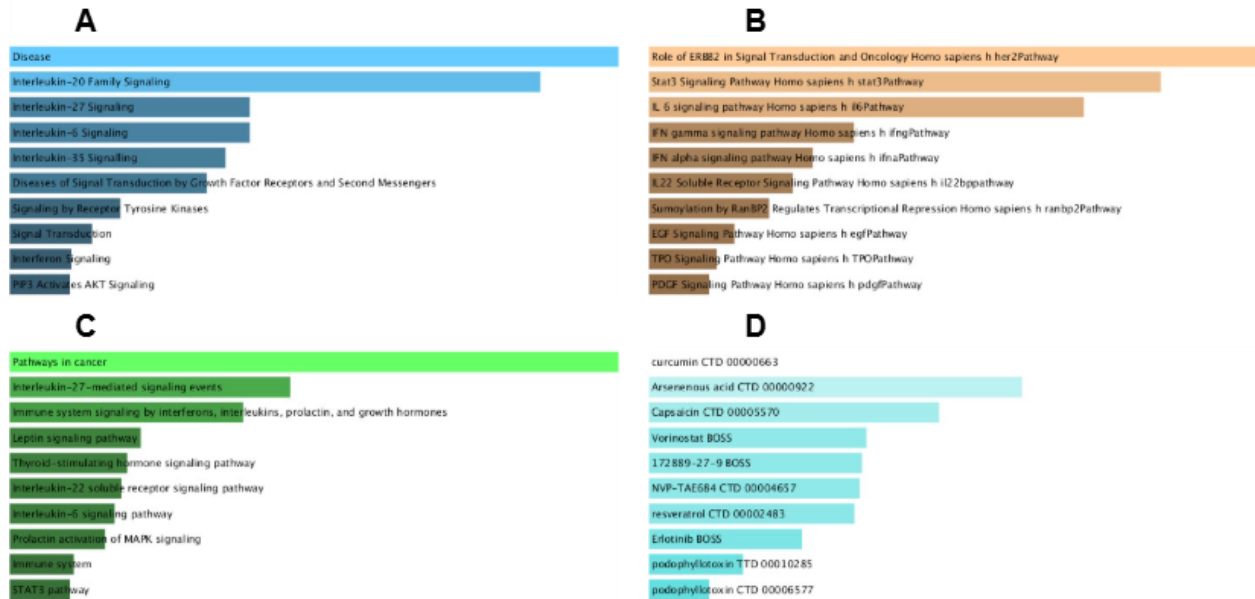


Fig. S5: Functional enrichment analysis of DEGs in depression and stress. The bar graphs illustrate significant pathways from (A) Reactome, (B) BioCarta, (C) BioPlanet, and (D) DsigBD databases, highlighting the main shared molecular mechanisms between depression and stress.

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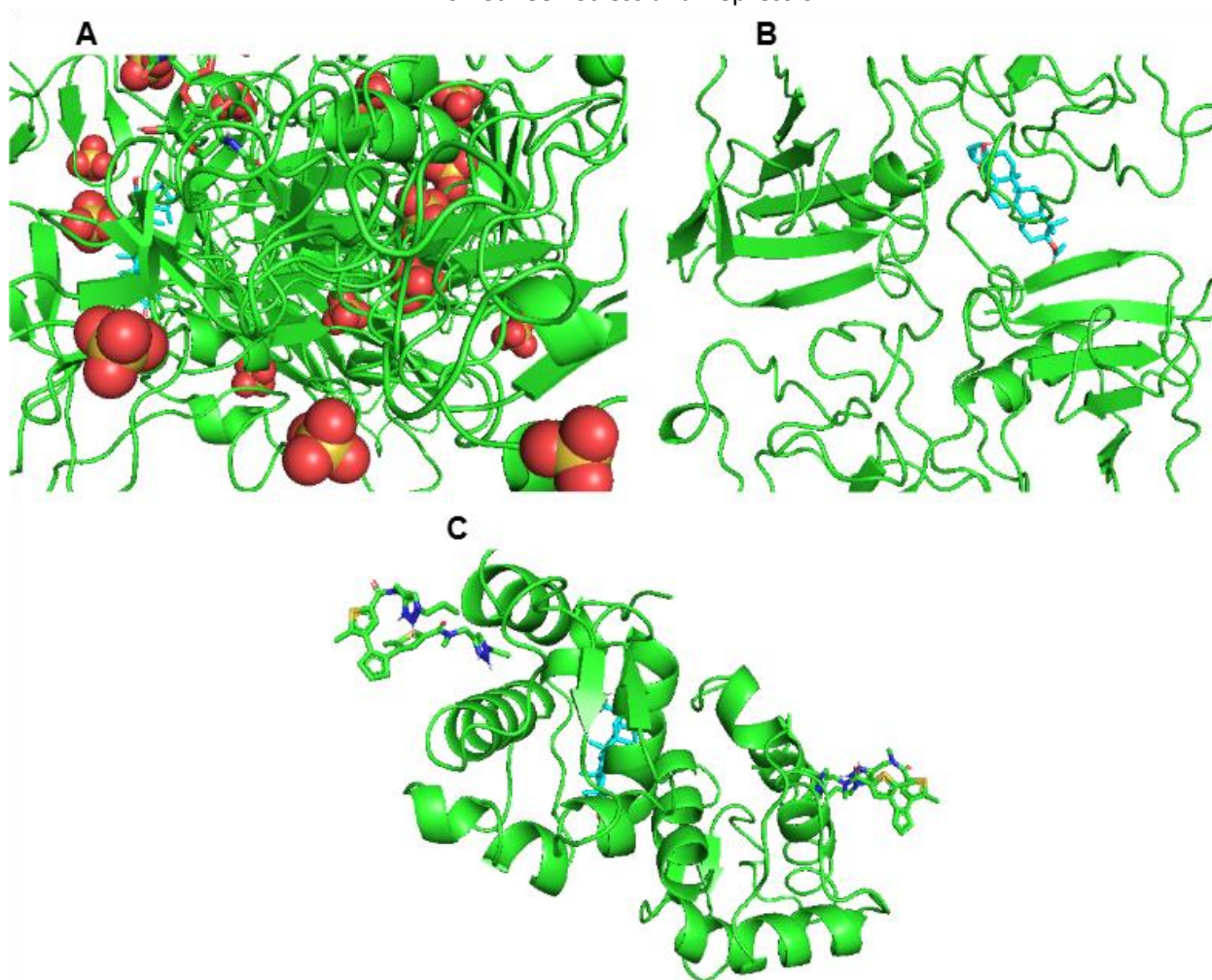


Fig. S6: Molecular docking configurations of the three selected genes with the drug compound 3-acetylursolic acid: (A) ERBB2, (B) HGF, and (C) MDM2. The target proteins are shown in blue, and the ligand (3-acetylursolic acid) is depicted in green, highlighting the binding pockets and conformational fit within each protein's active site.

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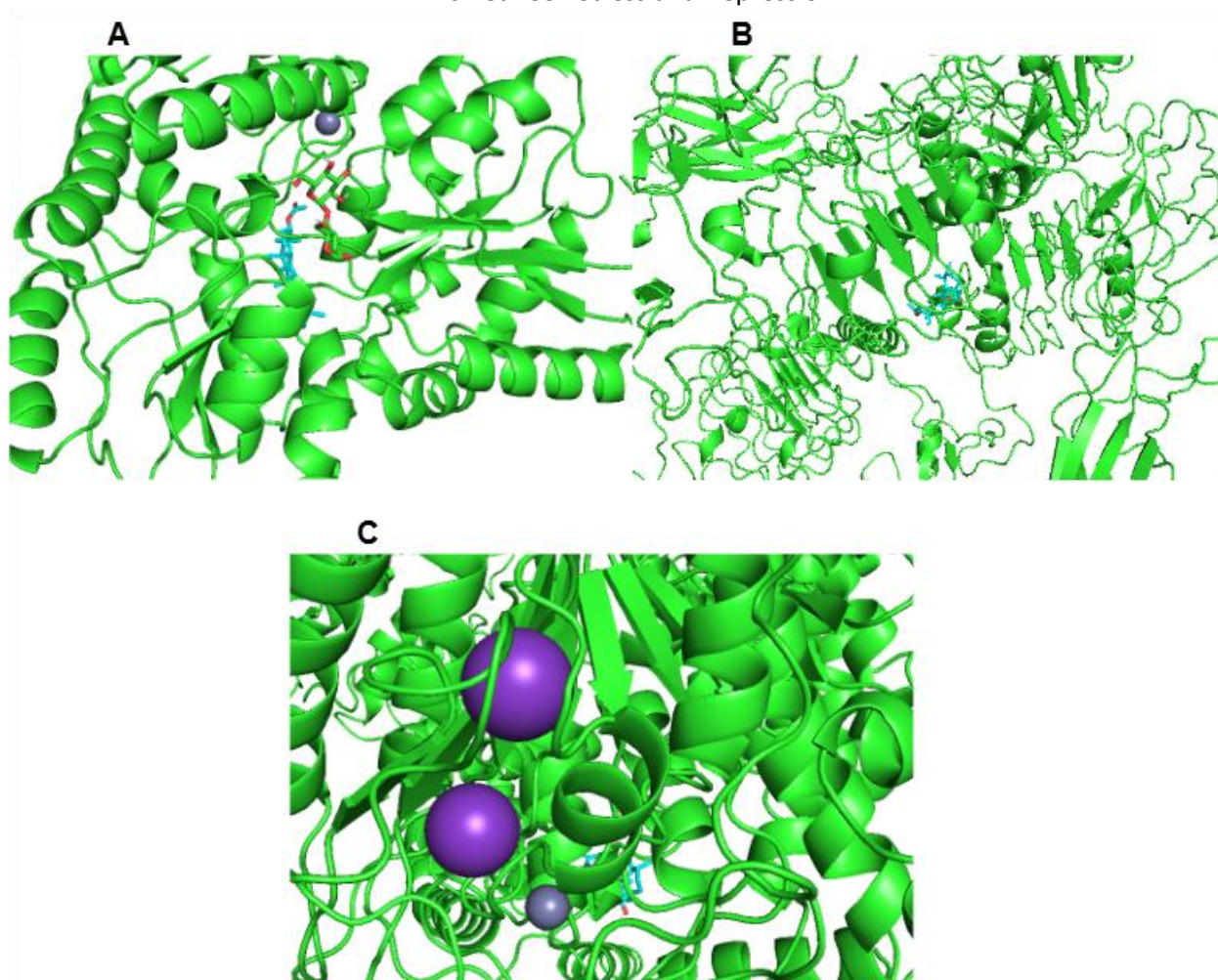


Fig. S7: Molecular docking configurations of the three selected genes with the drug compound 3-acetylursolic acid: (A) PML, (B) GF1R, and (C) HDAC1. The target proteins are shown in blue, and the ligand (3-acetylursolic acid) is depicted in green, highlighting the binding pockets and conformational fit within each protein's active site.

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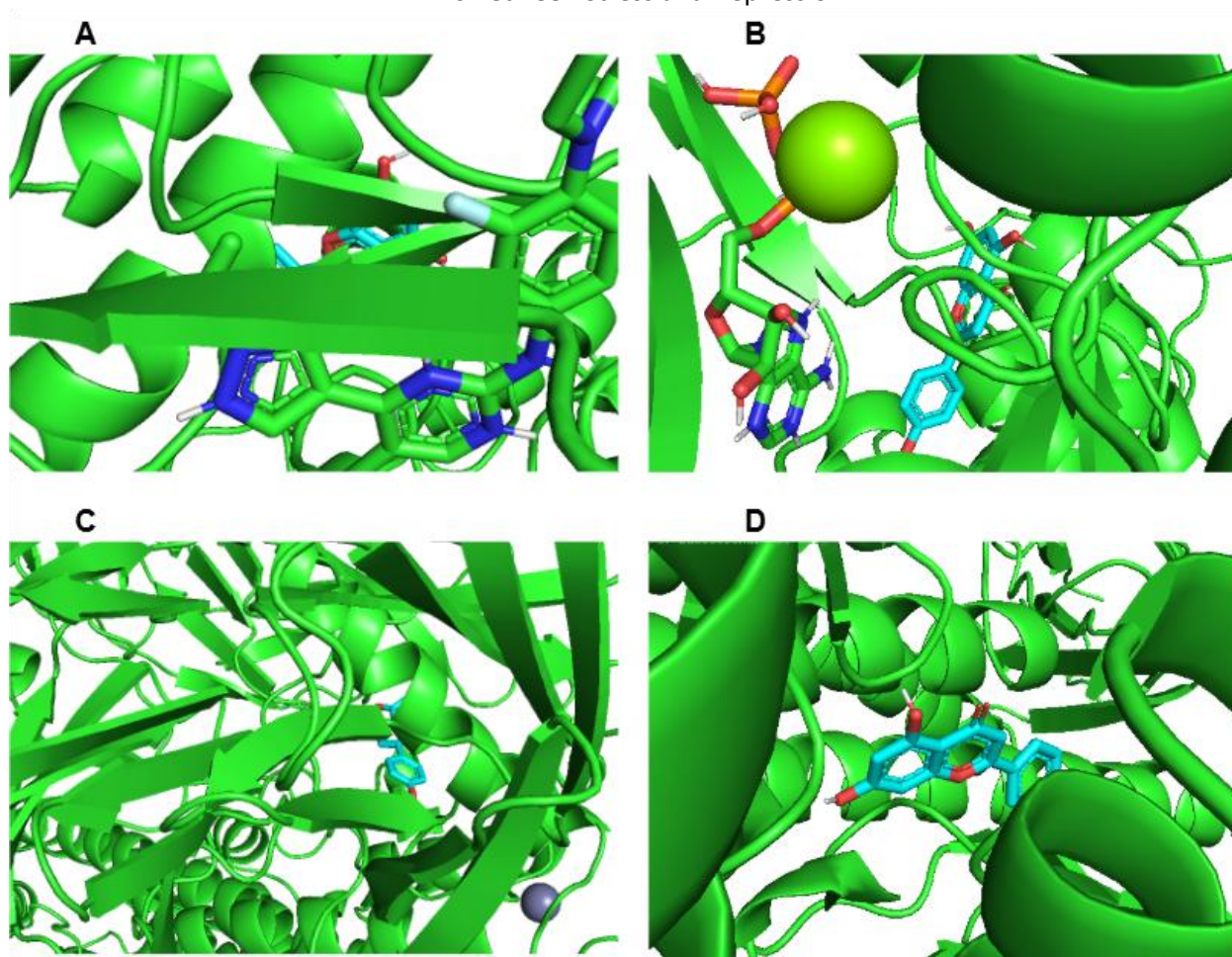


Fig. S8: Molecular docking representations of the four selected hub genes (A) JAK2, (B) JAK1, (C) STAT2, and (D) STAT3 with the drug compound Apigenin. The target proteins are shown in blue and the ligand (Apigenin) in green, illustrating the binding pockets and conformational fit in each protein's active site.

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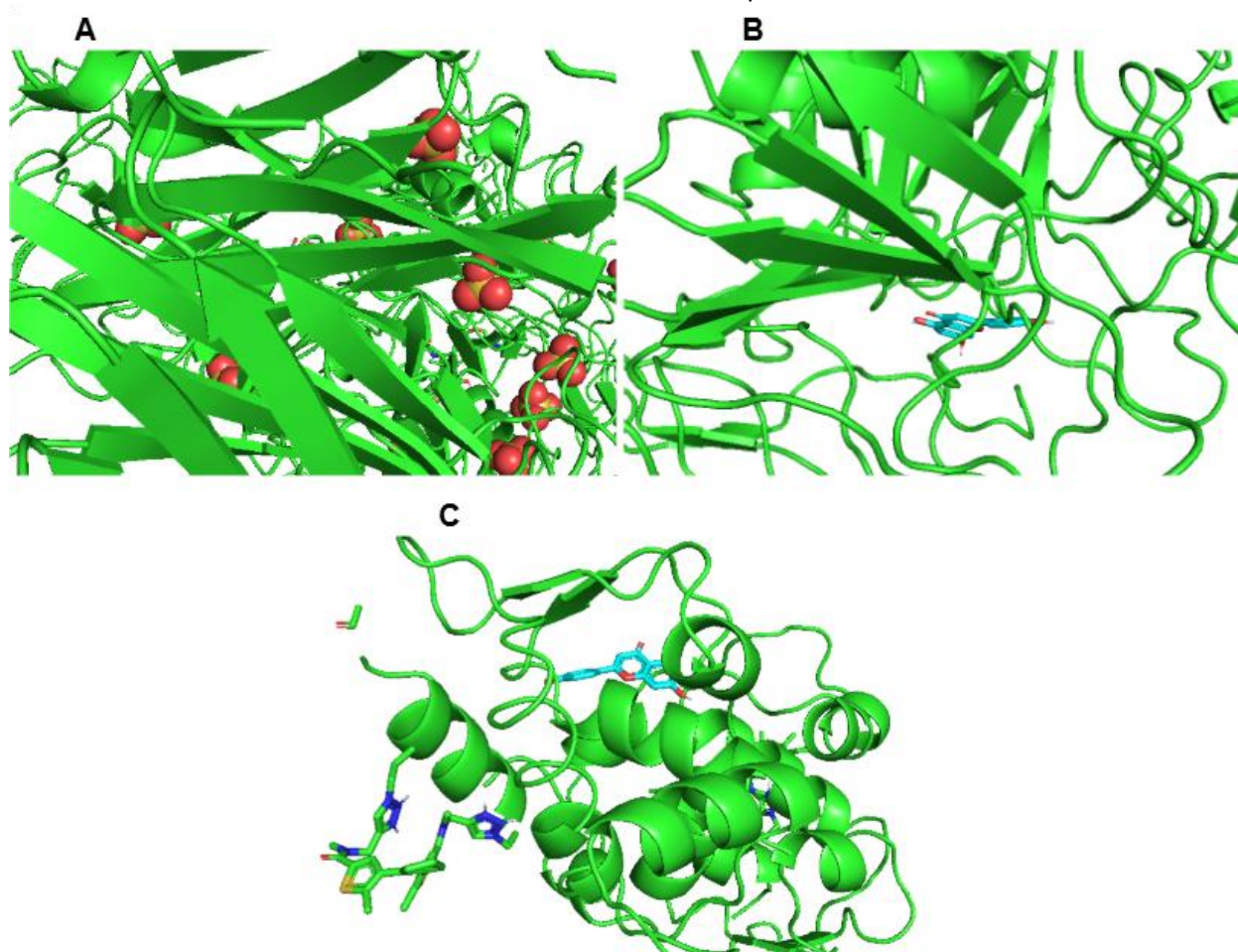


Fig. S9: Molecular docking configurations of the three selected genes with the drug compound Apigenin: (A) ERBB2, (B) HGF, and (C) MDM2. The target proteins are shown in blue, and the ligand is depicted in

Supplementary Files

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green, highlighting the binding pockets and conformational fit within each protein's activesite.

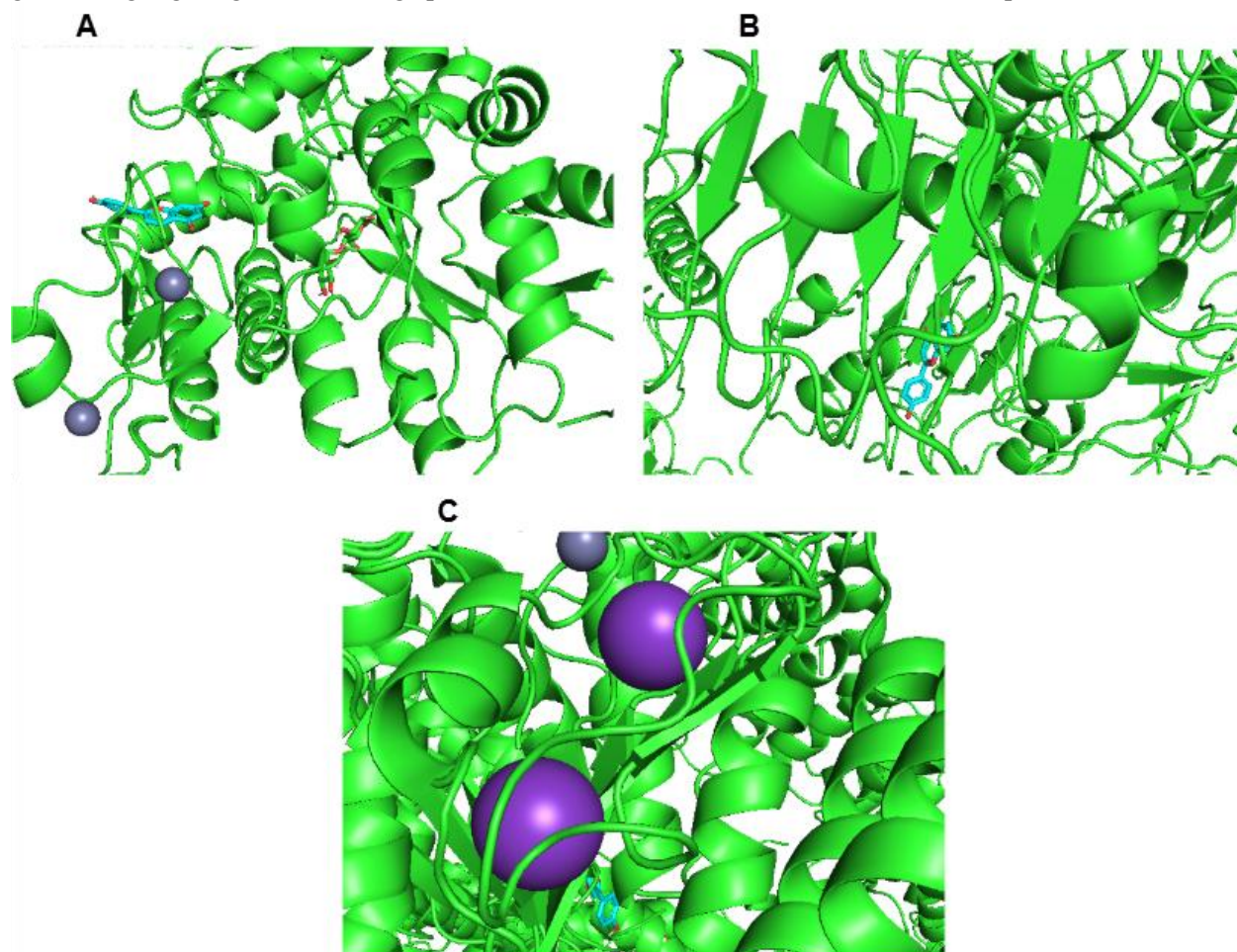


Fig. S10: Molecular docking configurations of the three selected genes with the drug compound Apigenin: (A) PML, (B) GF1R, and (C) HDAC1. The target proteins are shown in blue, and the ligand (Apigenin) is

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depicted in green, highlighting the binding pockets and conformational fit within each protein's active site.

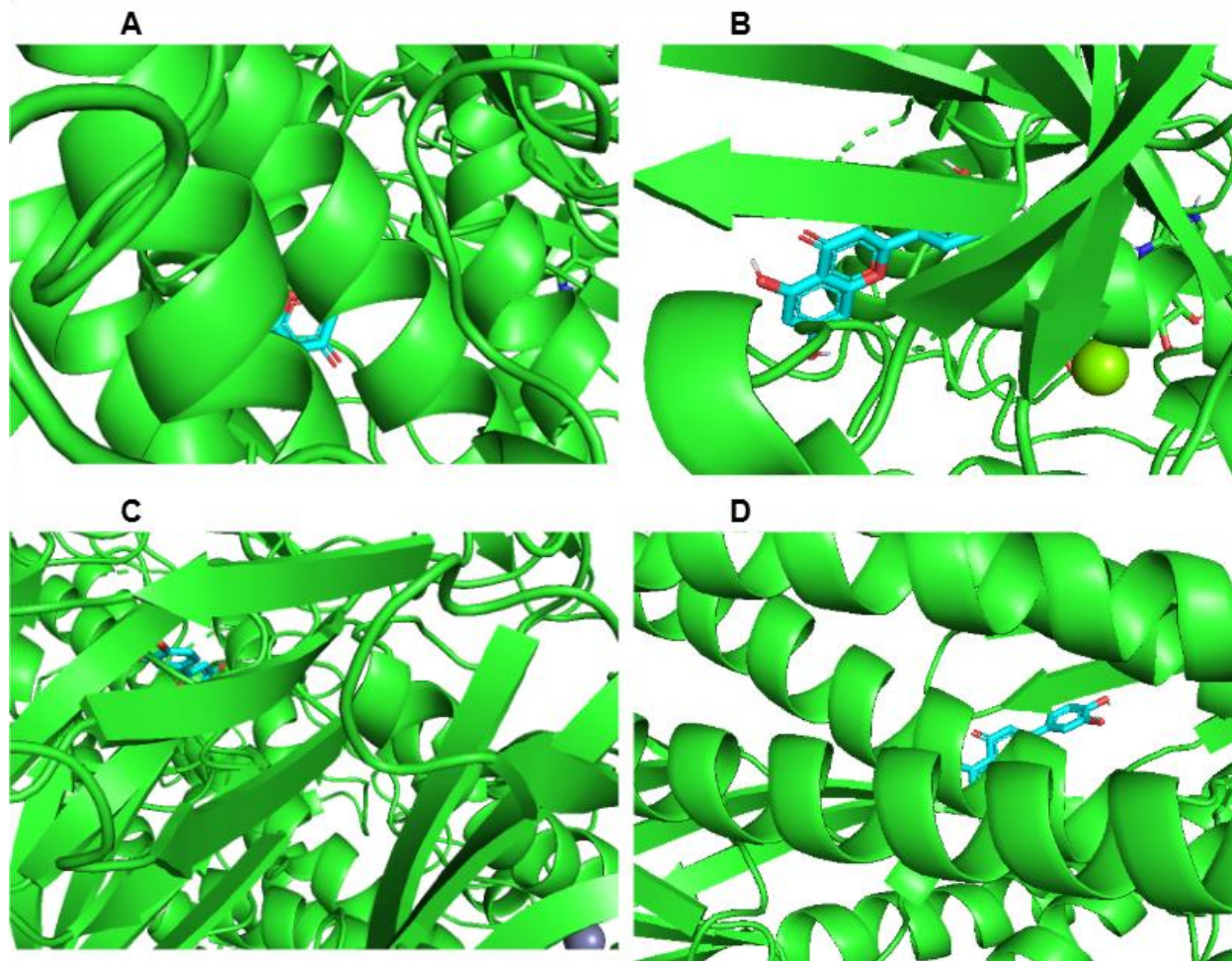


Fig. S11: Molecular docking representations of the four selected hub genes (A) JAK2, (B) JAK1, (C) STAT2, and (D) STAT3 with the drug compound Luteolin. The target proteins are shown in blue and the ligand (Luteolin) in green, illustrating the binding pockets and conformational fit in each protein's active site.

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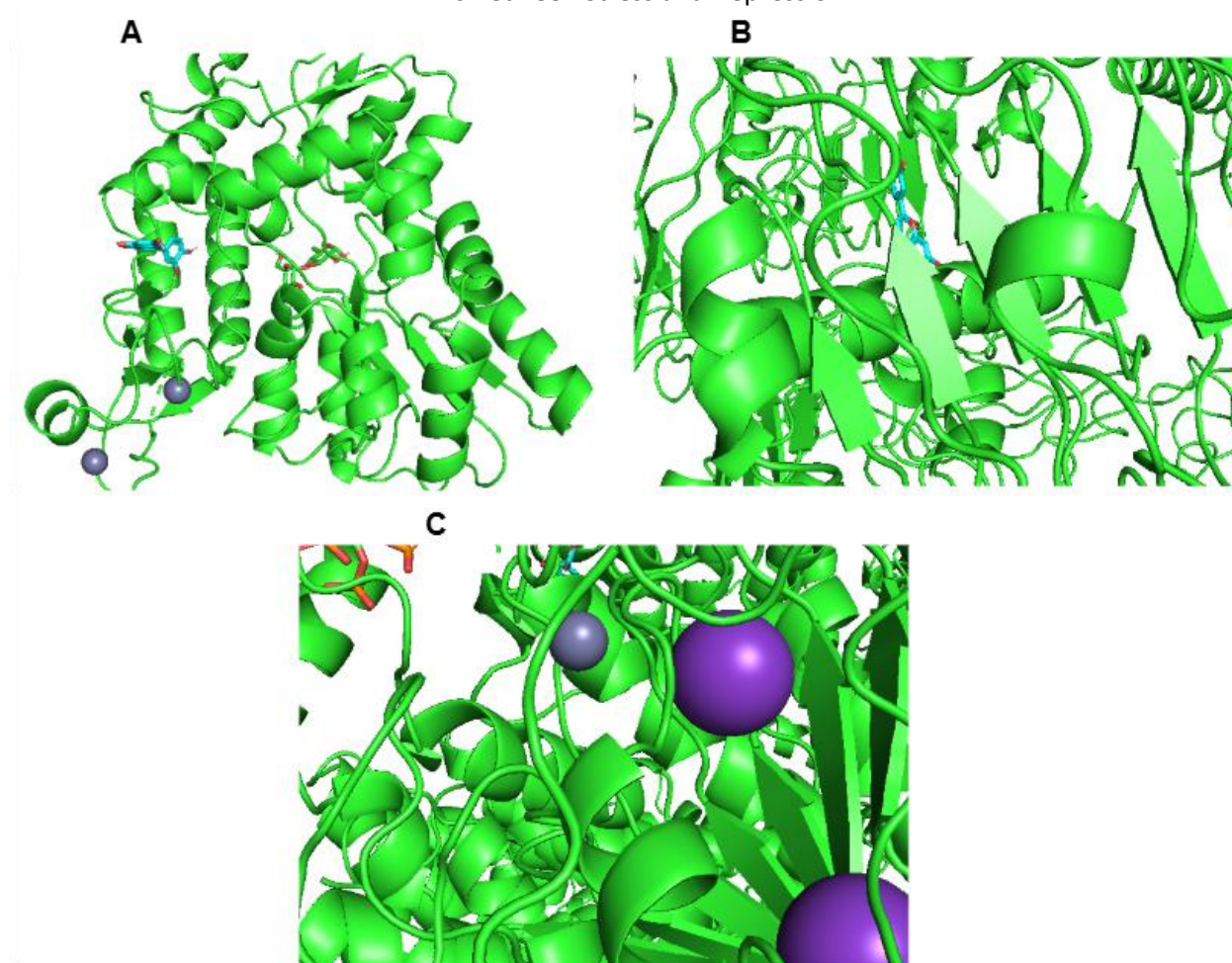


Fig. S12: Molecular docking configurations of the three selected genes with the drug compound Luteolin: (A) ERBB2, (B) HGF, and (C) MDM2. The target proteins are shown in blue, and the ligand (Luteolin) is depicted in green, highlighting the binding pockets and conformational fit within each protein's active site.

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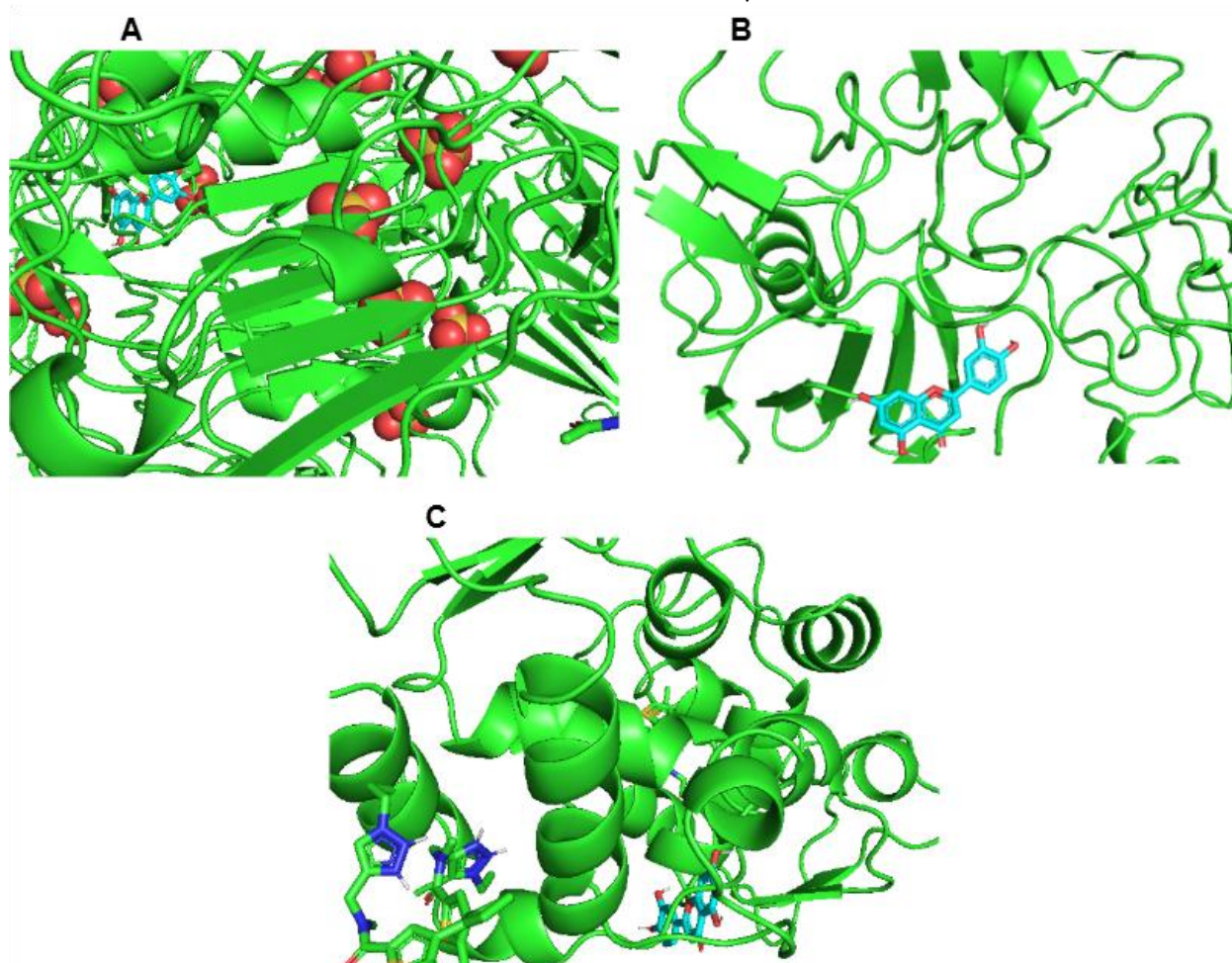


Fig. S13: Molecular docking configurations of the three selected genes with the drug compound Luteolin: (A) PML, (B) GF1R, and (C) HDAC1. The target proteins are shown in blue, and the ligand (Luteolin) is depicted in green, highlighting the binding pockets and conformational fit within each protein's active site.

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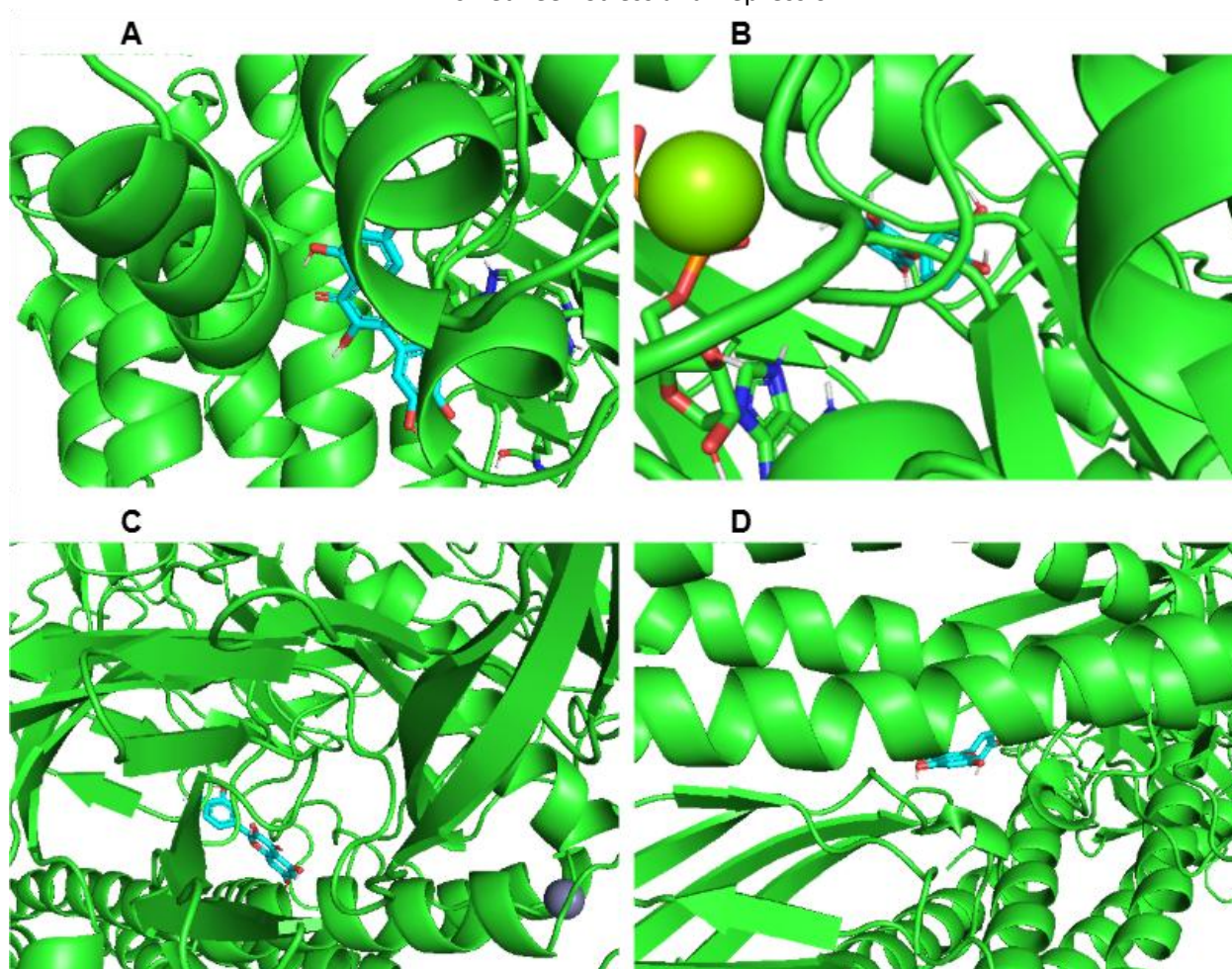


Fig. S14: Molecular docking representations of the four selected hub genes (A) JAK2, (B) JAK1, (C) STAT2, and (D) STAT3 with the drug compound Quercetin. The target proteins are shown in blue and the ligand (Quercetin) in green, illustrating the binding pockets and conformational fit in each protein's active site.

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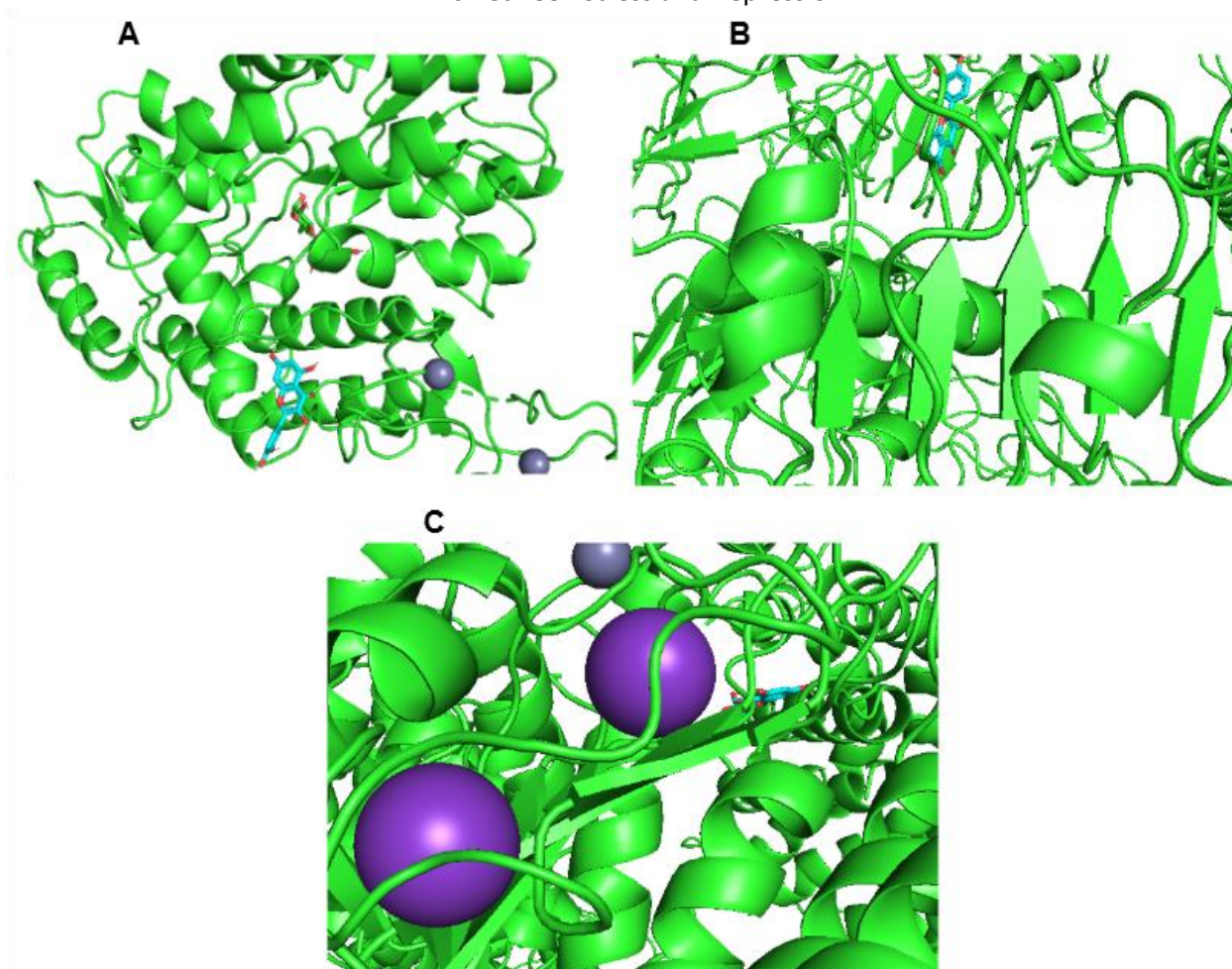


Fig. S15: Molecular docking configurations of the three selected genes with the drug compound Quercetin: (A) ERBB2, (B) HGF, and (C) MDM2. The target proteins are shown in blue, and the ligand (Quercetin) is depicted in green, highlighting the binding pockets and conformational fit within each protein's active site.

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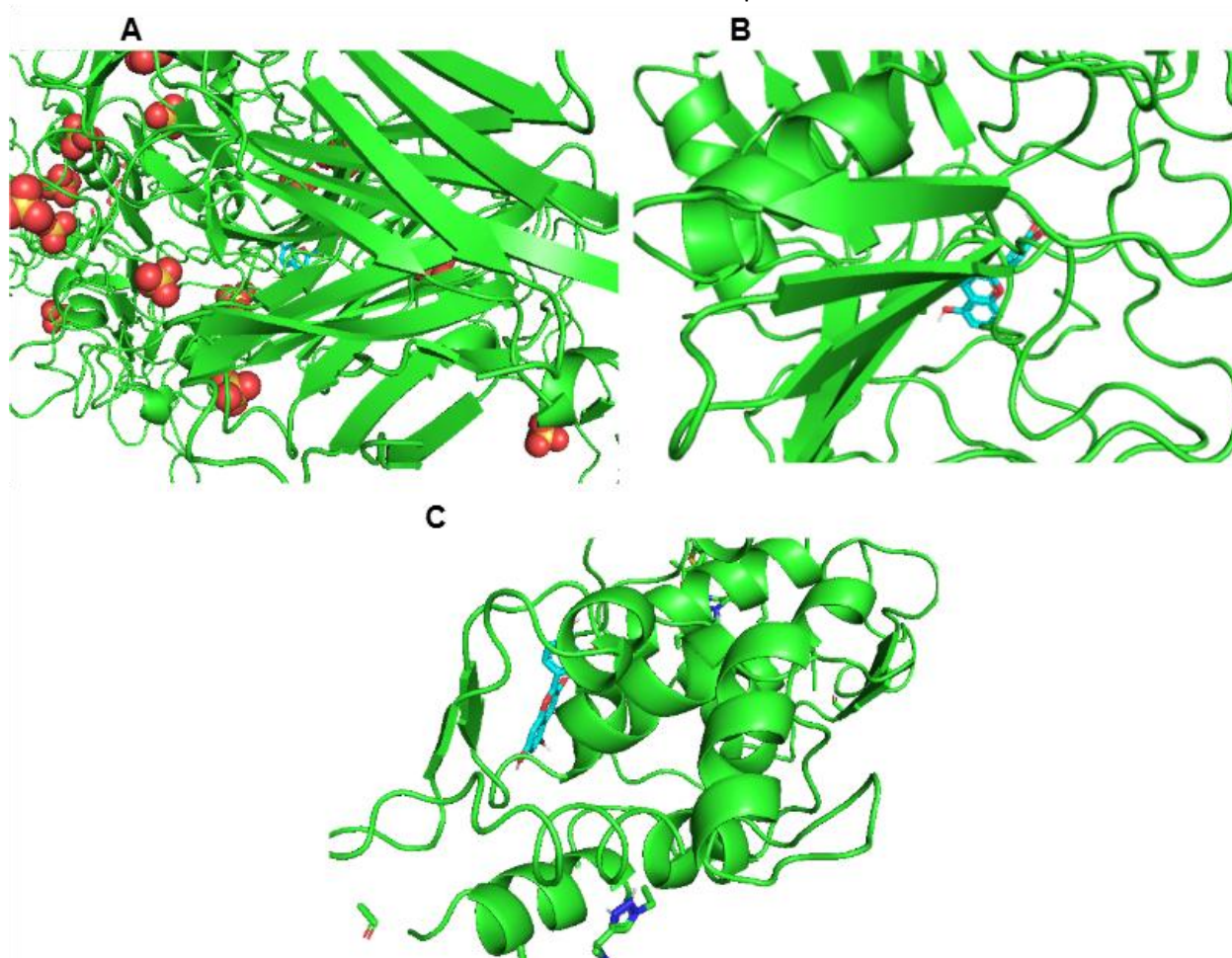


Fig. S16: Molecular docking configurations of the three selected genes with the drug compound Quercetin: (A) PML, (B) GF1R, and (C) HDAC1. The target proteins are shown in blue, and the ligand (Quercetin) is depicted in green, highlighting the binding pockets and conformational fit within each protein's active site.

GitHub Link: <https://github.com/MostaryKhatun/stress-depression-biomarker-discovery/tree/main>